3.4 Laplace transform pairs

Objective 3.4.1

Learn to use Laplace transform properties and pairs to compute the transform of a function.

The table below shows commonly-used Laplace transform pairs.

Table 3.4.1: Examples of Laplace transform pairs. $x(t) = 0$ for $t < 0^-$ and $T \geq 0$.

	$x(t)$	$\Longleftrightarrow$	$\mathbf{X}(\mathbf{s}) = \mathcal{L}[x(t)]$
1	$\delta(t)$	$\Longleftrightarrow$	1
1a	$\delta(t-T)$	$\Longleftrightarrow$	$e^{-T\mathbf{s}}$
2	$u(t)$	$\Longleftrightarrow$	$\frac{1}{\mathbf{s}}$
2a	$u(t-T)$	$\Longleftrightarrow$	$\frac{e^{-T\mathbf{s}}}{\mathbf{s}}$
3	$e^{-at}u(t)$	$\Longleftrightarrow$	$\frac{1}{\mathbf{s}+a}$
3a	$e^{-a(t-T)}u(t-T)$	$\Longleftrightarrow$	$\frac{e^{-T\mathbf{s}}}{\mathbf{s}+a}$
4	$t\,u(t)$	$\Longleftrightarrow$	$\frac{1}{\mathbf{s}^2}$
4a	$(t-T)u(t-T)$	$\Longleftrightarrow$	$\frac{e^{-T\mathbf{s}}}{\mathbf{s}^2}$

5	$t^2u(t)$	$\Longleftrightarrow$	$\frac{2}{\mathbf{s}^3}$
6	$te^{-at}u(t)$	$\Longleftrightarrow$	$\frac{1}{(\mathbf{s}+a)^2}$
7	$t^2e^{-at}u(t)$	$\Longleftrightarrow$	$\frac{2}{(\mathbf{s}+a)^3}$
8	$t^{n-1}e^{-at}u(t)$	$\Longleftrightarrow$	$\frac{(n-1)!}{(\mathbf{s}+a)^n}$
9	$\sin(\omega_0 t)u(t)$	$\Longleftrightarrow$	$\frac{\omega_0}{\mathbf{s}^2+{\omega_0}^2}$
10	$\sin(\omega_0 t+\theta)u(t)$	$\Longleftrightarrow$	$\frac{\mathbf{s}\cdot\sin\theta+\omega_0\cos\theta}{\mathbf{s}^2+{\omega_0}^2}$
11	$\cos(\omega_0 t)u(t)$	$\Longleftrightarrow$	$\frac{\mathbf{s}}{\mathbf{s}^2+{\omega_0}^2}$
12	$\cos(\omega_0 t+\theta)u(t)$	$\Longleftrightarrow$	$\frac{\mathbf{s}\cdot\cos\theta-\omega_0\sin\theta}{\mathbf{s}^2+{\omega_0}^2}$
13	$e^{-at}\sin(\omega_0 t)u(t)$	$\Longleftrightarrow$	$\frac{\omega_0}{(\mathbf{s}+a)^2+{\omega_0}^2}$
14	$e^{-at}\cos(\omega_0 t)u(t)$	$\Longleftrightarrow$	$\frac{\mathbf{s}+a}{(\mathbf{s}+a)^2+{\omega_0}^2}$
15	$2e^{-at}\cos(bt-\theta)u(t)$	$\Longleftrightarrow$	$\frac{e^{j\theta}}{\mathbf{s}+a+jb}+\frac{e^{-j\theta}}{\mathbf{s}+a-jb}$
15a	$e^{-at}\cos(bt-\theta)u(t)$	$\Longleftrightarrow$	$\frac{(\mathbf{s}+a)\cos\theta+b\sin\theta}{(\mathbf{s}+a)^2+b^2}$
16	$\frac{2t^{n-1}}{(n-1)!}e^{-at}\cos(bt-\theta)u(t)$	$\Longleftrightarrow$	$\frac{e^{j\theta}}{(\mathbf{s}+a+jb)^n}+\frac{e^{-j\theta}}{(\mathbf{s}+a-jl}$